

Requirement Analysis phase-II

4. Technology Stack

Technology Stack Overview

Alternative Stack Options

Option A (100% ServiceNow OOB): Use Predictive Intelligence instead of Python Flask - no external hosting needed but requires premium license. Option B (Power Platform): Power Automate Flow + AI Builder + Dataverse for organizations using Microsoft stack. Option C (Open Source): Jira + n8n Flow + Python FastAPI + HuggingFace Transformers for startups.

Date	28.09.2026
Team ID	SWTID-2026-6482
Project Name	Auto Ticket Classification using Flow Designer

Tool	Purpose
ServiceNow Platform (Incident Table), Service Portal, Virtual Agent	Ticket creation interface, user portal
Flow Designer, Business Rules, REST API Step, Predictive Intelligence	No-code automation, trigger, API integration
Python (Flask, Scikit-learn, Pandas, NLTK, TF-IDF + Logistic Regression / Naive Bayes, Optional: BERT / DistilBERT), Flask API	Train model, predict Category/Priority
Databases (sys_journal, incident, sys_user_group), Historical Tickets CSV (Kaggle IT Tickets 10k), Feedback Table	Training data, storing predictions
Cloud (AWS Lambda, Azure Functions), PDI (Personal Developer Instance), Python Flask hosted on Render / HuggingFace / Local, GitHub for versioning	Deployment, hosting, CI/CD
Reporting (Grafana), Dashboards, Performance Analytics, Email Notifications	Accuracy tracking, SLA monitoring